THE UNIVERSAL BOUNDARY ANNIHILATION THEOREM

Exact Collapse and Annihilation of Left Diagonals in Elementary Cellular Automaton Rule 30

The Setting & Rowland's Open Bound:
Stephen Wolfram's Rule 30 is defined by $c(t+1,x) = c(t,x-1) \oplus (c(t,x) \vee c(t,x+1))$.
Its left diagonals $D_{j(t)} = c(t, j-t)$ follow the Rowland-Wolfram recurrence:

$$D_{j+2}(t+1) = D_{j(t)} \oplus \big(D_{j+1}(t) \vee D_{j+2}(t)\big)$$

The Rowland (2006) upper bound states that diagonal j has period dividing 2^j ($2^5 = 32$, $2^7 = 128$). We replace this loose existential bound with two exact closed-form collapse theorems:

Master Ladder Theorems (Lean 4 Certified, Zero Custom Axioms)

(I) Diagonal 5 Collapse:
$\boldsymbol{D_5(t) = D_3(t)} \quad (\forall t \geq 5)$
Exact period 2, shattering $2^5 = 32$.

(II) Diagonal 7 Annihilation ($z_2 = 7$):
$\boldsymbol{D_7(t) = 0} \quad (\forall t \geq 7)$
Identically zero, shattering $2^7 = 128$.

The Complete Constructive Proof:

1. **Boundary Seeds**: $D_0(t) = 1$ ($\forall t \geq 0$) and $D_1(t) = 1$ ($\forall t \geq 1$).
2. **First Zero Diagonal ($z_1 = 2$)**: Setting $j = 0$ in the recurrence:

   $$D_2(t+1) = D_0(t) \oplus (D_1(t) \vee D_2(t)) = 1 \oplus (1 \vee D_2) = 1 \oplus 1 = \mathbf{0} \quad (\forall t \geq 1).$$

   Hence $D_2(t) = 0$ for all $t \geq 2$.
3. **Strict Alternation of Diagonal 3**: Setting $j = 1$ and substituting $D_2 = 0$:

   $$D_3(t+1) = D_1(t) \oplus (D_2(t) \vee D_3(t)) = 1 \oplus (0 \vee D_3) = 1 \oplus D_3 = \boldsymbol{\neg D_3(t)} \quad (\forall t \geq 2).$$

   Hence D_3 strictly alternates with period 2 ($D_3(t+2) = D_3(t)$).
4. **Unit Absorption of Diagonal 4**: Setting $j = 2$ and substituting $D_2 = 0$:

   $$D_4(t+1) = D_2(t) \oplus (D_3(t) \vee D_4(t)) = 0 \oplus (D_3 \vee D_4) = D_3(t) \vee D_4(t).$$

   Since $D_3(3) = 1$, $D_4(4) = 1$, and an OR accumulator with 1 is locked: $\boldsymbol{D_4(t) = 1}$ for all $t \geq 4$.
5. **Collapse of Diagonal 5**: Setting $j = 3$ for $t \geq 4$ and substituting $D_4(t) = 1$:

   $$D_5(t+1) = D_3(t) \oplus (D_4(t) \vee D_5(t)) = D_3(t) \oplus (1 \vee D_5) = D_3(t) \oplus 1 = \neg D_3(t) = \boldsymbol{D_3(t+1)}.$$

   Integrating yields $\boldsymbol{D_5(t) = D_3(t)}$ for all $t \geq 5$. $\blacksquare_1$
6. **Collapse of Diagonal 6**: Setting $j = 4$ for $t \geq 4$: $D_6(t+1) = D_4(t) \oplus (D_5(t) \vee D_6(t)) = \neg(D_5(t) \vee D_6(t))$. By induction from $t = 6$ ($D_6(6) = D_5(6) = 0$), $D_5 \vee D_6 = D_5$, so $D_6(t+1) = \neg D_5(t) = D_5(t+1)$. Thus $\boldsymbol{D_6(t) = D_5(t)}$ for all $t \geq 6$.
7. **The Annihilation of Diagonal 7 ($x \oplus x = 0$ Cancellation)**: Setting $j = 5$:

   $$D_7(t+1) = D_5(t) \oplus (D_6(t) \vee D_7(t))$$

   By induction from $t = 7$ ($D_7(7) = 0$): substituting $D_7(t) = 0$ and $D_6(t) = D_5(t)$ yields:

   $$D_7(t+1) = D_5(t) \oplus (D_5(t) \vee 0) = D_5(t) \oplus D_5(t) = \mathbf{0} \quad (\forall t \geq 7). \quad \blacksquare_2$$

Author: Umidjon Rasulov (School of Computer and Information Engineering, Inha University in Tashkent).
Official Verification: Formalized in Lean 4 (`Rule30Submission.original_diagonal_invariants`).
Kernel verified via `WolframInstitute/Rule30PrizeKit` (`Check.lean`) with exit code 0, standard axioms only (`propext`, `Quot.sound`).